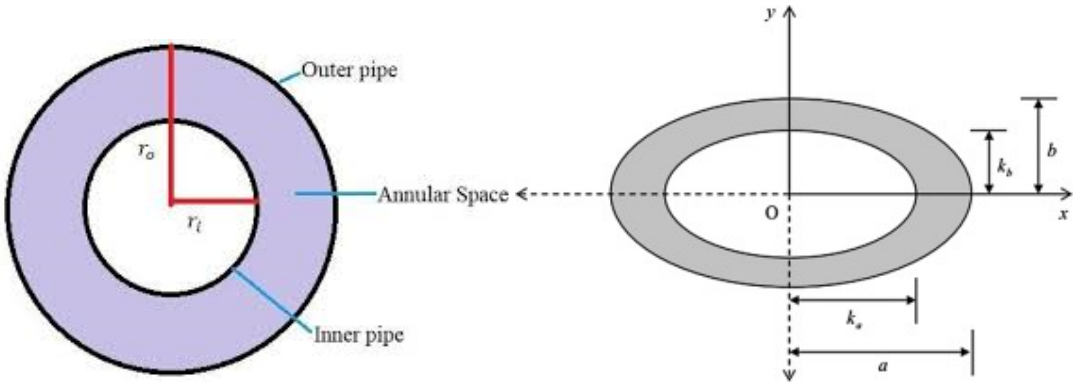
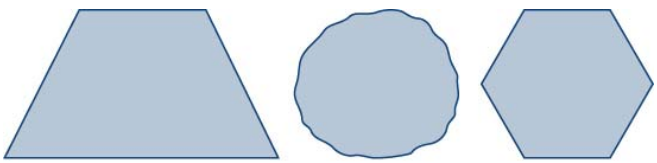
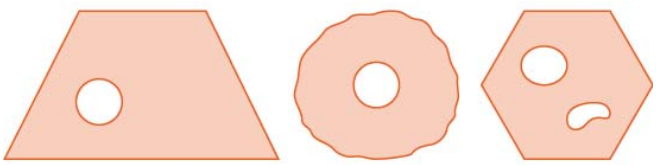
An overview of various multiply-connected regions

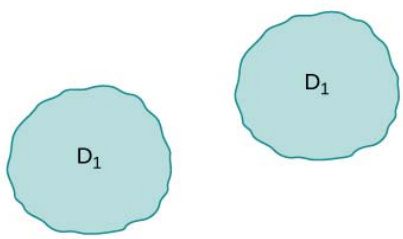




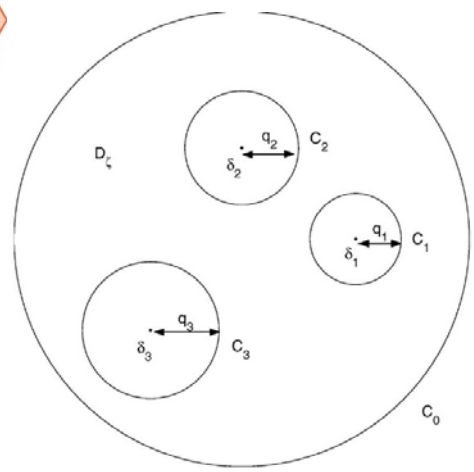
(a) Simply connected regions

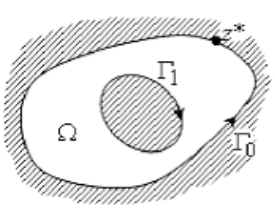


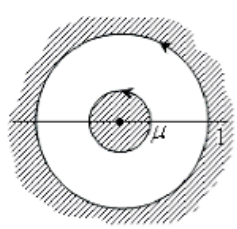
(b) Connected regions that are not simply connected

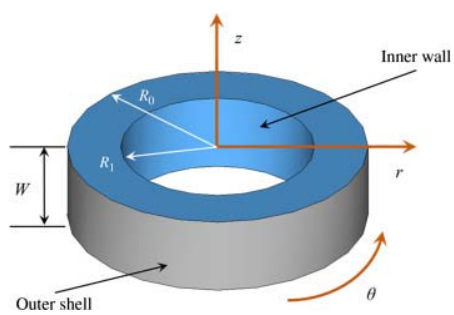


(c) A region that is not connected









Multiply connected region in marine hydrodynamics

In **marine hydrodynamics**, a **multiply connected region** occurs when the fluid domain contains one or more "holes" — obstacles, cavities, or excluded regions — so that a closed curve cannot always be shrunk to a point without leaving the fluid. These situations are extremely common in real ocean engineering.

General mathematical structure

For a multiply connected marine flow region:

$$\nabla \times q = 0 \quad \text{in the fluid}$$

but

$$\oint_C q \cdot dr = \Gamma \neq 0$$

for loops enclosing holes.

The velocity potential becomes **multi-valued**:

$$\phi = \text{single-valued part} + \Gamma\theta$$

where θ is the angular coordinate around the obstacle.

This explains:

- existence of vortices
- lift generation
- logarithmic singularities
- circulation conservation

Stokes' theorem in a multiply connected region

1. Statement of Stokes' theorem

For a smooth vector field **q** over a surface S with boundary C,

$$\oint_C \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS$$

This relates **circulation around the boundary** to **vorticity over the surface**.

2. Multiply connected region: geometry

Consider a 2D fluid region with a hole (an obstacle).

Let:

- C_0 = outer boundary (counterclockwise)
- C_1 = inner boundary around the hole (clockwise)
- S = surface between them

The **total boundary** is:

$$\partial S = C_0 - C_1$$

(negative sign because orientations must be consistent with the normal).

3. Applying Stokes' theorem

Stokes' theorem applies to the **entire boundary**:

$$\oint_{C_0} \mathbf{q} \cdot d\mathbf{r} - \oint_{C_1} \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS$$

This is the key point:

👉 **Stokes' theorem does not fail in multiply connected regions**

👉 It automatically includes *all boundary components*

4. Special case: irrotational flow

Suppose

$$\nabla \times \mathbf{q} = 0 \quad \text{in } S$$

Then

$$\iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS = 0$$

So Stokes gives:

$$\oint_{C_0} \mathbf{q} \cdot d\mathbf{r} = \oint_{C_1} \mathbf{q} \cdot d\mathbf{r}$$

Interpretation

The circulations around all boundaries are equal.

Thus circulation can be **non-zero**, even though curl = 0 everywhere in the fluid.

This is the fundamental property of multiply connected domains.

Why simply connected case is different

If the region had **no hole**, we could shrink the loop to a point.

Then Stokes would imply:

$$\oint_C \mathbf{q} \cdot d\mathbf{r} = 0$$

But the hole prevents contraction.

Hence circulation survives.

Final conclusion

In a multiply connected region:

$$\oint_{\text{all boundaries}} \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS$$

For irrotational flow:

$$\oint_{C_0} = \oint_{C_1} = \Gamma$$

Stokes theorem is not violated — it **enforces equality of circulations** around different boundaries.

This is the mathematical foundation of:

- vortex dynamics
- hydrofoil lift
- circulation theory in marine hydrodynamics

1. From simply connected → multiply connected

In a **simply connected region**:

Any closed curve can shrink to a point.

Applying Stokes' theorem:

$$\oint_C \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS$$

If the flow is irrotational:

$$\nabla \times \mathbf{q} = 0$$

then

$$\oint_C \mathbf{q} \cdot d\mathbf{r} = 0$$

So circulation must vanish.

2. Region with multiple structures

Suppose the fluid contains **N solid obstacles**.

The domain has:

- outer boundary C_0
- inner boundaries $C_1, C_2, \ldots, C_N$

Each inner boundary surrounds one structure.

The total boundary is:

$$\partial S = C_0 - \sum_{k=1}^{N} C_k$$

(negative orientation for inner loops).

3. Extended Stokes theorem (key result)

Applying Stokes to the entire region:

$$\oint_{C_0} \mathbf{q} \cdot d\mathbf{r} - \sum_{k=1}^{N} \oint_{C_k} \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS$$

Irrotational hydrodynamic flow

If

$$\nabla \times \mathbf{q} = 0$$

then

$$\oint_{C_0} = \sum_{k=1}^{N} \oint_{C_k}$$

Interpretation

The outer circulation equals the **sum of circulations around all structures**.

Each structure can carry its own circulation:

$$\Gamma_k = \oint_{C_k} \mathbf{q} \cdot d\mathbf{r}$$

Thus

$$\Gamma_{\text{outer}} = \sum_{k=1}^{N} \Gamma_k$$

This is the extension from simple to multiply connected regions.

4. Hydrodynamic meaning

In marine hydrodynamics:

Each structure acts as a **topological hole** that can support independent circulation.

Examples:

- vortices around offshore piles
- lift on multiple hydrofoils
- wake circulation behind ship arrays

The velocity potential becomes:

$$\phi = \phi_0 + \sum_{k=1}^{N} \frac{\Gamma_k}{2\pi} \theta_k$$

where:

- θ_k = angular coordinate around structure k
- each term is multi-valued

Stokes' theorem guarantees **global consistency** of these circulations.

Final generalized statement

For a region with N holes:

$$\boxed{\sum_{\text{all boundary components}} \oint \mathbf{q} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{q}) \cdot \mathbf{n}\, dS}$$

For irrotational marine flow:

$$\boxed{\Gamma_0 = \sum_{k=1}^{N} \Gamma_k}$$

This is how Stokes theorem extends from simply connected to multiply connected domains in hydrodynamics.

Multivalued functions in multiply connected region

1. Physical definition of circulation

In two–dimensional flow with velocity field

$$\mathbf{q} = (u(x,y),\, v(x,y)),$$

the circulation around a closed contour C is defined as

$$\boxed{\Gamma(C) = \oint_C \mathbf{q} \cdot d\mathbf{r} = \oint_C (u\, dx + v\, dy)}$$

This is the fundamental hydrodynamic definition.

2. Complex representation of velocity

Introduce the complex variable

$$z = x + iy.$$

In potential flow we define the **complex velocity potential**

$$\boxed{W(z) = \phi(x,y) + i\psi(x,y)}$$

where

- ϕ = velocity potential
- ψ = stream function

They satisfy the Cauchy–Riemann equations:

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \qquad v = \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}.$$

The complex velocity is

$$\boxed{\frac{dW}{dz} = u - iv}$$

3. Express the line element in complex form

The differential of z is

$$dz = dx + i\,dy.$$

Multiply complex velocity by dz:

$$(u - iv)(dx + i\,dy).$$

Expand:

$$(u - iv)(dx + i\,dy) = u\,dx + iu\,dy - iv\,dx + v\,dy.$$

Group real and imaginary parts:

$$= (u\,dx + v\,dy) + i(u\,dy - v\,dx).$$

4. Identification of circulation

The **real part** of this expression is exactly the circulation integrand:

$$u\,dx + v\,dy = \mathrm{Re}\left[\frac{dW}{dz}\,dz\right].$$

Therefore

$$\Gamma(C) = \oint_C (u\,dx + v\,dy) = \mathrm{Re} \oint_C \frac{dW}{dz}\,dz.$$

But

$$\frac{dW}{dz}\,dz = dW.$$

Hence

General circulation formula

$$\boxed{\Gamma(C) = \mathrm{Re} \oint_C dW = \mathrm{Re}\left[\Delta W(C)\right]}$$

5. Interpretation via potential and stream function

Since

$$dW = d\phi + i\,d\psi,$$

we have

$$\oint_C dW = \oint_C d\phi + i \oint_C d\psi.$$

Thus

$$\boxed{\Gamma(C) = \oint_C d\phi}$$

because circulation equals the net change in velocity potential.

6. Simply connected region case

If $W(z)$ is single-valued:

$$\oint_C dW = 0$$

so

$$\Gamma(C) = 0.$$

Thus irrotational flow in a simply connected region has zero circulation.

7. Multiply connected region (multivalued potential)

If $W(z)$ is multivalued, then after one circuit:

$$W \to W + \Delta W.$$

Hence

$$\boxed{\Gamma(C) = \mathrm{Re}[\Delta W]}$$

For example, if

$$W(z) = \frac{\Gamma}{2\pi i} \log z,$$

then after one loop:

$$\log z \to \log z + 2\pi i,$$

so

$$\Delta W = \Gamma.$$

Thus

$$\Gamma(C) = \Gamma.$$

Circulation equals the **jump in the multivalued complex potential**.

8. Final compact general statement

The most general definition of circulation in complex potential theory is:

$$\boxed{\Gamma(C) = \mathrm{Re} \oint_C dW = \mathrm{Re} \oint_C \frac{dW}{dz}\, dz = \mathrm{Re}[\Delta W(C)]}$$

where:

- If W is single-valued → circulation = 0
- If W is multivalued → circulation equals its branch jump
- In multiply connected regions, circulation is topological

Theorem: General decomposition of a multivalued analytic potential

Statement

Let D be a multiply connected region in the complex plane with holes at

$$a_1, a_2, \ldots, a_N.$$

If $W(z)$ is an analytic (possibly multivalued) complex potential in D, then it can be written in the form

$$\boxed{W(z) = F(z) + \sum_{k=1}^{N} A_k \log(z - a_k)}$$

where

- $F(z)$ is **single-valued analytic** in D
- A_k are constants
- logarithmic terms contain **all multivaluedness**

1. Key idea of the proof

The essential observation is:

👉 Multivaluedness comes from integrating functions with **simple poles**

because

$$\int \frac{dz}{z - a} = \log(z - a).$$

Thus we study the derivative

$$G(z) = \frac{dW}{dz}.$$

Even if $W(z)$ is multivalued, **its derivative is single-valued**.

2. Structure of the derivative

Since $G(z)$ is analytic in D except possibly at isolated singularities a_k, it admits a decomposition

$$G(z) = H(z) + \sum_{k=1}^{N} \frac{A_k}{z - a_k},$$

where

- $H(z)$ is analytic everywhere in D
- the second term contains all simple pole singularities

This follows from standard complex analysis: any meromorphic function can be written as an analytic part plus principal parts.

3. Integration of the derivative

Now integrate:

$$W(z) = \int G(z)\,dz = \int H(z)\,dz + \sum_{k=1}^{N} A_k \int \frac{dz}{z - a_k}.$$

Define

$$F(z) = \int H(z)\,dz,$$

which is **single-valued analytic** in D.

And

$$\int \frac{dz}{z - a_k} = \log(z - a_k).$$

Thus

$$W(z) = F(z) + \sum_{k=1}^{N} A_k \log(z - a_k).$$

4. Why logarithms create multivaluedness

When a contour circles a_k once,

$$\log(z - a_k) \to \log(z - a_k) + 2\pi i.$$

Hence

$$W(z) \to W(z) + 2\pi i A_k.$$

This jump produces circulation and multivalued behavior.

The function $F(z)$ contributes **no jump** because it is single-valued.

5. Physical hydrodynamic interpretation

In fluid mechanics we write

$$A_k = \frac{\Gamma_k}{2\pi i}.$$

Then

$$W(z) = F(z) + \sum_{k=1}^{N} \frac{\Gamma_k}{2\pi i} \log(z - a_k).$$

Here:

- Γ_k = vortex strength
- each hole behaves like a concentrated vortex
- circulation is encoded in logarithmic branches

6. Why this decomposition is general

This result is general because:

- Any analytic multivaluedness arises from integrating singular terms
- Only simple poles produce logarithmic branches
- Higher-order poles give single-valued algebraic terms
- Essential singularities are excluded in physical flows

Thus logarithms represent the **complete source of multivaluedness**.

General formula for circulation in a multiply connected region

1. General circulation formula

From complex potential theory, circulation around a closed contour C is

$$\boxed{\Gamma(C) = \mathrm{Re} \oint_C dW = \mathrm{Re} \oint_C \frac{dW}{dz}\, dz}$$

where $W(z)$ is the complex velocity potential.

2. Structure of a multivalued potential in a multiply connected region

Consider a region with N holes (excluded points)

$$a_1, a_2, \ldots, a_N.$$

In such a region, the most general analytic multivalued potential can be decomposed as

$$\boxed{W(z) = F(z) + \sum_{k=1}^{N} A_k \log(z - a_k)}$$

where:

- $F(z)$ = single-valued analytic function
- A_k = complex constants
- logarithmic terms generate multivaluedness

Different circuits around holes change the logarithm by

$$\log(z - a_k) \to \log(z - a_k) + 2\pi i\, n_k,$$

where n_k is the **winding number** of contour C about a_k.

3. Differentiate the potential

$$\frac{dW}{dz} = F'(z) + \sum_{k=1}^{N} \frac{A_k}{z - a_k}.$$

Substitute into the circulation formula:

$$\Gamma(C) = \mathrm{Re} \oint_C \left[F'(z) + \sum_{k=1}^{N} \frac{A_k}{z - a_k} \right] dz.$$

4. Contribution of the single-valued part

Since $F'(z)$ is analytic inside the region,

$$\oint_C F'(z)\,dz = 0$$

by Cauchy's theorem.

Thus only logarithmic singularities contribute.

5. Evaluate singular contributions

For each hole,

$$\oint_C \frac{dz}{z - a_k} = 2\pi i\, n_k(C),$$

where $n_k(C)$ is the winding number.

Therefore

$$\Gamma(C) = \mathrm{Re}\left[\sum_{k=1}^{N} A_k (2\pi i\, n_k)\right].$$

6. Physical interpretation of constants

For hydrodynamic circulation, we write

$$A_k = \frac{\Gamma_k}{2\pi i},$$

where Γ_k is the vortex strength at the k-th hole.

Substituting:

$$\Gamma(C) = \mathrm{Re}\sum_{k=1}^{N} \frac{\Gamma_k}{2\pi i}(2\pi i\, n_k).$$

Hence we obtain the **general circulation formula**:

General formula in multiply connected region

$$\boxed{\Gamma(C) = \sum_{k=1}^{N} \Gamma_k\, n_k(C)}$$

7. Equivalent derivation via jump in multivalued function

Since

$$\oint_C dW = \Delta W(C),$$

and each logarithm contributes a jump

$$A_k(2\pi i n_k),$$

we have

$$\Delta W(C) = \sum_{k=1}^{N} A_k(2\pi i n_k).$$

Taking the real part again yields

$$\Gamma(C) = \sum_{k=1}^{N} \Gamma_k n_k.$$

Thus circulation equals the **total branch jump** of the multivalued potential.

8. Final theorem (general statement)

Theorem — Circulation in a multiply connected region

If the complex potential is

$$W(z) = F(z) + \sum_{k=1}^{N} \frac{\Gamma_k}{2\pi i} \log(z - a_k),$$

then the circulation around any closed contour C is

$$\boxed{\Gamma(C) = \sum_{k=1}^{N} \Gamma_k \, n_k(C)}$$

where $n_k(C)$ are winding numbers.

9. Deep meaning

This result shows:

- Circulation is a **topological invariant**
- It depends only on how many times the contour wraps around holes
- Local irrotationality does not force zero circulation
- Multivaluedness of $W(z)$ encodes vortex strengths

Are multivalue functions natural or mathematical curiosities?

Multivalued functions naturally and unavoidably arise in real physical problems in marine hydrodynamics and ocean engineering. They are not mathematical curiosities. They encode essential topological and physical features of fluid motion, especially in circulation and vortex dynamics.

Characteristics of multivalued functions

- It is single-valued for each specific loop
- But different homotopy classes of loops give different values
- These values arise from multivalued velocity potentials
- Zero vorticity does not imply zero circulation

This is why vortex motion and hydrofoil flows naturally involve **multiply connected domains** and **nonzero circulation**.

If the fluid domain is **multiply connected** (contains obstacles, vortices, or holes), then:

👉 zero vorticity does **not** imply zero circulation
👉 the velocity potential cannot remain single-valued

To represent nonzero circulation mathematically, the potential must be **multivalued**, typically through logarithmic terms.

Thus:

Multivalued potentials are the mathematical expression of physical circulation.

Role of winding number in circulation

From the general circulation formula:

$$\boxed{\Gamma(C) = \sum_k \Gamma_k \, n(C, a_k)}$$

This means:

👉 the winding number acts as a **multiplier of vortex strength**

Physically:

- Each vortex contributes circulation
- The contribution is weighted by how many times the loop surrounds it

Thus winding number determines measurable flow properties.

Example: Whether opposite vortices cancel each other.
Consider

$$W(z) = \frac{\Gamma}{2\pi i}\log(z-a) - \frac{\Gamma}{2\pi i}\log(z-b).$$

Loop enclosing both vortices

$$\Gamma(C) = \Gamma - \Gamma = 0.$$

Loop enclosing only one vortex

$$\Gamma(C) = \pm\Gamma.$$

Vortex motion in marine hydrodynamics

Examples in nature:

- whirlpools in tidal channels
- vortex shedding behind ships
- eddies in ocean currents
- propeller tip vortices

Circulation around hydrofoils and ship hulls

Lift generation on hydrofoils and ship rudders requires **nonzero circulation**.

In hydrofoil theory:

- the trailing edge creates a multiply connected domain
- the complex potential includes a logarithmic term
- circulation determines lift via the **Kutta–Joukowski theorem**

$$L = \rho U \Gamma.$$

Without multivalued potentials, lift cannot be represented.

Engineering applications:

- ship propeller design
- rudder hydrodynamics
- underwater vehicle control surfaces
- wave–structure interaction

Flow around offshore structures

Vortex shedding and circulation around these structures involve:

- multivalued potentials
- topological circulation constants
- vortex-induced vibrations

These are critical in:

- fatigue analysis
- structural stability
- coastal engineering

Wave–vortex interaction in ocean engineering

In coastal and ocean flows:

- eddies interact with surface waves
- circulation modifies wave scattering
- multivalued potentials describe trapped modes

Applications include:

- harbor resonance
- sediment transport
- coastal protection
- tsunami and storm surge modeling

Ocean eddies and large-scale circulation

Ocean eddies behave like giant vortices:

- they trap water masses
- conserve circulation
- form multiply connected domains

Applications:

- climate modeling
- pollutant transport
- sediment dynamics
- offshore navigation

Multivalued potentials describe conserved circulation in geophysical flows.

Summary table

Mathematical concept	Physical meaning	Engineering relevance
Logarithmic branch	Circulation/vortex	Lift and propulsion
Winding number	Topological loop class	Flow connectivity
Multivalued potential	Persistent vortex structure	Wake dynamics
Branch jump	Conserved circulation	Stability analysis

Example 1: Two offshore piles in irrotational current

Problem

Two cylindrical offshore piles stand in an irrotational ocean current.

Measured circulations:

$$\Gamma_1 = 4\ \mathrm{m}^2/\mathrm{s}, \quad \Gamma_2 = -1.5\ \mathrm{m}^2/\mathrm{s}$$

Find the circulation around a large loop enclosing both piles.

Solution

By Stokes theorem in multiply connected domain:

$$\Gamma_{\text{outer}} = \Gamma_1 + \Gamma_2$$

So:

$$\Gamma_{\text{outer}} = 4 + (-1.5) = 2.5\,\mathrm{m}^2/\mathrm{s}$$

Interpretation

The far-field flow carries net circulation **2.5 m²/s**.

This affects:

- wake interaction
- vortex shedding forces
- fatigue loading

Example 2: Annular flow around a submerged structure

Problem

Velocity field in an annular region:

$$\mathbf{q} = \left(-\frac{3y}{x^2+y^2}, \frac{3x}{x^2+y^2}\right)$$

Find circulation around a circle of radius 5 m.

Solution

Switch to polar coordinates:

$$\mathbf{q} = \frac{3}{r}\mathbf{e}_\theta$$

Line element:

$$d\mathbf{r} = r\,d\theta\,\mathbf{e}_\theta$$

Dot product:

$$\mathbf{q} \cdot d\mathbf{r} = 3\,d\theta$$

Circulation:

$$\Gamma = \int_0^{2\pi} 3\,d\theta = 6\pi$$

Key point

Circulation is **independent of loop size** — direct consequence of Stokes theorem in a multiply connected region.

Example 3: Three-structure harbor system

Problem

Three harbor structures generate circulations:

$$\Gamma_1 = 2.2, \quad \Gamma_2 = -0.7, \quad \Gamma_3 = 1.5$$

If a survey ship measures zero circulation far away, check consistency.

Solution

Total inner circulation:

$$\Gamma_{\text{total}} = 2.2 - 0.7 + 1.5 = 3.0$$

But measured outer circulation = 0.

This contradicts Stokes theorem:

$$\Gamma_{\text{outer}} = \sum \Gamma_k$$

So measurements are inconsistent.

At least one circulation value must be wrong.

Engineering meaning

This is used in:

- CFD validation
- experimental flow diagnostics
- vortex balance checks

Example 4: Vortex strength from measured velocity

Problem

A current meter records tangential velocity:

$$v_\theta = 0.4\,\mathrm{m/s}$$

at radius $r = 2$ m around a monopile.

Find circulation.

Solution

$$\Gamma = \oint v_\theta\, ds = \int_0^{2\pi} (0.4)(2)\, d\theta$$

$$\Gamma = 0.8 \times 2\pi = 1.6\pi \approx 5.03\,\mathrm{m^2/s}$$

Interpretation

This circulation must equal circulation around **any loop** enclosing the pile.

Summary of numerical lessons

All examples show:

$$\Gamma_{\text{outer}} = \sum \Gamma_{\text{inner}}$$

This is Stokes theorem extended to marine multiply connected domains.

It enables engineers to:

- ☑ check vortex conservation
- ☑ validate flow measurements
- ☑ model multi-body interactions
- ☑ predict wake behavior

Question

An idealized 2D ocean current near a coastal structure is modeled by the velocity field

$$\mathbf{q} = (u, v) = (ky, 0)$$

where $k = 0.4\ \mathrm{s}^{-1}$.

A rectangular loop in the water has corners at
$(0,0), (4,0), (4,3), (0,3)$ meters.

1. Find the circulation around the rectangle using a line integral.
2. Verify the result using Stokes' theorem.

Solution

Step 1 — Circulation by line integral

Circulation:

$$\Gamma = \oint_C \mathbf{q} \cdot d\mathbf{r}$$

Break into four sides.

Side 1: bottom edge (y = 0)

Here $u = k(0) = 0$

$$\Gamma_1 = 0$$

Side 2: right edge (x = 4)

Motion vertical:

$$d\mathbf{r} = (0, dy), \quad \mathbf{q} = (ky, 0)$$

Dot product = 0

$$\Gamma_2 = 0$$

Side 3: top edge (y = 3)

$$u = k(3) = 1.2$$

Direction is leftward:

$$\Gamma_3 = \int_4^0 1.2\, dx = -4.8$$

Side 4: left edge

Dot product = 0

$$\Gamma_4 = 0$$

Total circulation

$$\Gamma = -4.8\ \mathrm{m}^2/s$$

(Sign depends on orientation.)

Magnitude:

$$|\Gamma| = 4.8$$

Step 2 — Using Stokes' theorem

Curl of velocity:

$$\nabla \times \mathbf{q} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0 - k = -0.4$$

Area of rectangle:

$$A = 4 \times 3 = 12$$

Stokes theorem:

$$\Gamma = \iint_S (-0.4)\, dA = -0.4 \times 12 = -4.8$$

Final answer

$$\boxed{\Gamma = -4.8\ \mathrm{m}^2/s}$$

Magnitude of circulation:

$$\boxed{|\Gamma| = 4.8\ \mathrm{m}^2/s}$$

Physical interpretation

The ocean current has **uniform shear**, producing constant vorticity.

Circulation equals:

vorticity × enclosed area

This is a direct engineering application of Stokes' theorem in coastal flows.

Q1.

State Stokes' theorem and explain how it applies to a **multiply connected fluid region** containing offshore structures. Show mathematically why circulation can be non-zero even if the flow is irrotational.

Q2.

Define **circulation** in fluid mechanics. Explain its physical meaning in ocean engineering with one example involving a vortex around a monopile.

Q3.

Distinguish between:

- Simply connected region
- Multiply connected region

Explain their importance in marine hydrodynamics and vortex motion.

Q4.

Show that for a 2D velocity field

$$\mathbf{q} = (u, v)$$

circulation around a closed curve equals the surface integral of vorticity over the enclosed area.